



Determining Optimal Lag Time Selection Function with Novel Machine Learning Strategies for Better Agricultural Commodity Prices Forecasting in Malaysia

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ABSTRACT

In 2018, agriculture remains an important sector of Malaysia's economy, contributing around 7.54% to national GDP and 11.09% to the total employment. However, in agriculture, price volatility is often unpredictable due to the reliability of agricultural production on natural phenomena. Instability of prices endanger the development of Malaysia's economy and the food accessibility by consumers, which may lead to food insecurity, hunger and malnutrition. Thus, the need of an accurate forecasting model for agricultural commodity price is acute, especially for government and farmers to propose new policies and better plantation plan to deal with potential risks in the markets. In the context of forecasting, optimal lag selection is important to improve the forecast performance in terms of time and accuracy. The main goal of this research is to study and design novel machine learning strategies with optimal lag time selection function to forecast agricultural commodity price more accurately in order to improve plantation plan in Malaysia.

CCS Concepts

•Computing methodologies→ Neural networks •Information systems → Data analytics •Computing methodologies → Supervised learning by regression• Computing methodologies →Feature selection

Keywords

Time Series Analysis; Price Forecasting; Neural Networks; Agriculture; Differential Evolution

1. INTRODUCTION

Agriculture sector includes not only the production of essential food crops, but also production and industry beyond farming, such as bee keeping, forestry, poultry and dairy farming. For many years, agriculture formed the basis of economy system in Malaysia. Hence, the role of agriculture in the development of national economy cannot be underestimated.

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Agriculture sector is known as the major contributors to national incomes and export earnings in the early development stage of Malaysian economy. Even though the contribution to Malaysia's GDP in the early 21st century has declined to roughly 10%.[4], agriculture remains an important sectors of Malaysia's economy. In 2018, agriculture contributed around 7.54 percent to national GDP [25] and 11.09 percent to the total employment [24]. However, there are many uncertainties of agriculture inherent in natural phenomena, weather, the handling of living organism and other aspects, which may lead to price volatility.[11] The instability of price can cause supply volatility, endanger the food accessibility and market trades.[29] Hence, policy-makers and other participants in the market are concerned with the determinants of price volatility. A time series analysis is also highly desired for them to better understand the expected future events and mitigate the potential agricultural risks.

Time series analysis has been heavily exploited in a variety of fields, especially in economics. Among the traditional approaches, the most frequently used statistical-based model is Autoregressive Integrated Moving Average (ARIMA). The drawback of this method is it assumes the relationship between the input and output is linear. Linear model may lead to under-fitting issues when dealing with more complicated data. Hence, nonlinear model such as SVM and ANN have been proposed. SVM have been widely applied in agriculture, such as automatic classifier to commercialize underutilized crops [37][39] and rice yield forecasting [31]. With recent rapid development in computing power, time series forecasting based on Artificial Neural Networks (ANNs) has been popular. ANNs have been proven to out-perform the classical statistical methods [16][17] and can be used to model nonlinear time series.[20]

A common question in time series analysis is the number of past data (lags) that is appropriate for the forecasting model. Various lag time selection methods have been proposed, for example AIC, HQC and BIC. However, these approaches cannot reserve the correctness when dealing with nonlinear data. Therefore, some nonlinear techniques have been proposed to fill the gap, such as Genetic Algorithms (GA), Particle Swarm Optimization (PSO) and Frankenstein's Particle Swarm Optimization (FPSO). In this research, a biological-inspired algorithm named as differential evolution is proposed to perform optimal lag time selection.

In this research, the main goal is to investigate price determinants in agricultural markets and design machine learning strategies to forecast commodity prices in Malaysia. To achieve higher forecast accuracy, differential evolution could be employed for optimal lag selection by discarding irrelevant or redundant time lags. Being able to accurately forecast the agricultural prices, government can

formulate new policies to intervene the potential situations occurs in the agricultural market. Meanwhile, farmers can make better decision on the plantation plan to improve the crop productivity.

2. LITERATURE REVIEW

2.1 Factors Influencing Agricultural Commodity Price

Many factors can cause price volatility of agricultural commodity. Commodity price depends on the interaction between the demand and supply of the agricultural market. Hence, price determinants can be categorised into demand-side factors and supply side factors.[2]

The role of crude oil in agriculture become increasingly important owing to the fact that crude oil is one of the in-put materials used for machinery and transportation. Thus, changes in crude oil prices may substantially impact the agricultural market. In recent years, many studies suggest that the impact of crude oil price on agricultural commodity price is significant.[33][14]

Due to the reliability of agricultural production on natural phenomena, adverse changes in climate and weather condition may affect to many aspects in agriculture, such as crops, livestock and fishery. For instance, temperature change can lead to livestock disease as well as quantity and quality of feeds, substantially affect the animal production, reproduction and health[8]. In literature, many studies have been carried out to investigate the relationship between climate or temperature change and agricultural commodity price. In [1], the analysis result shows that expected climate condition, rainfall has significant negative impact on both commodity spot price and future price. An extended version of storage models was proposed by [30] and the results find decreasing yield and increasing yield variability caused by climate change results in the increase of commodity price level and volatility.

Exchange rate is one of the demand-side factors in agriculture sector. The movement in exchange rate will influence agricultural import and exports, agricultural production and investment. In literature, there are studies suggest there is linkage between exchange rate and commodity price [13] while there are also studies suggest weak dependency between them. [10]

In the literature, studies on determinants of agricultural commodity prices in Malaysia are thin. There are works such as [15] and [12] aim to investigate the linkage between food price, oil price, GDP and exchange rate. Both research works applied a nonlinear autoregressive distributed lags (NARDL) model and found evidence to support the existence of relationship.

Factors determining the agricultural commodity prices deserve an attention from all the parties involved in Malaysia. Early preparation such as propose new control policies and make adjustment to the plantation plan to cope with all the potential situations that will impact the agricultural market.

2.2 Time Series Forecasting

Previous research studies on agricultural commodity prices forecasting model can be categorized into three groups, namely statistical model, intelligent model and hybrid models. Statistical approaches are popular and frequently used in time series forecasting at an early stage. Among all the traditional approaches, ARIMA have been widely adopted in the domain of agriculture for forecasting, such as cotton [5], onion [38] and rice [23].

However, it becomes increasingly clear that statistical approaches are not sufficient to deal with nonlinear data due to the complexity, which inspiring more research to study on machine learning methods in forecasting to fill the gap. In literature, there are many works have proven that ANN out-performed ARIMA in agricultural price forecasting[16][17]. There are also studies implemented hybrid model to further improve the performance of neural network models by handling both linear and nonlinear patterns.[35][21] Moreover, agricultural commodities prices are affected by many other factors. The forecasting model will be less reliable if the influential factors are not considered and this lead to studies that takes exogenous factors into the forecasting process.[26][32]

In Malaysia, previous work on agricultural commodity price forecasting mainly focused on the price of palm oil [18] and natural rubber [34][19]. These works demonstrated the success of using ARIMA model to forecast the price of palm oil and natural rubber. Besides, authors in [18] also find that the stock of palm oil and the crude petroleum oil price are have a significant impact on the crude palm oil. In summary, forecasts can be used as guidance for public use or private use. In other words, the need of an accurate and effective forecasting method is indeed vital to complement decision making on plantation or economics plan.

2.3 Optimal Lag Time Selection Function

In the context of time series forecasting, nonlinear models such as ANNs assume the past data have an important influence on the next data. Ideally, a small number of lag periods is desired which can unveil the unique features embedded in the data. In other words, the goal of the optimal lag time selection process is to obtain a proper set of lag that has critical influence with all the irrelevant or useless lag values discarded from the data. Too few or too many lags selected will impact the training performance of forecasting model.

Various classical approaches based on criteria for lag selection have been applied to forecasting especially in the economics domain, such as AIC, BIC and ACC. However, these approaches are not suitable for nonlinear model due to the nature of complexity inherent in time series.

In machine learning, there is a variety of methodologies exist for feature selection or lag selection. These methods can be divided into filters and wrappers.[3] Other alternative methods for feature selection are Particle Swarm Optimization (PSO) [7] and Frankenstein's PSO (FPSO) [27] and evolutionary algorithms [22]. FPSO is a method that combines components of several algorithms that made modifications in the original PSO. In [27], the authors find that FPSO performed better than PSO in all the experiments except one.

Another possible approach for lag selection is Differential Evolution (DE) which is first introduced by [28]. It is a branched of evolutionary computing which imitates the natural selection process. Due to the promising performance of DE in other domains [6], there is a potential to apply this technique for optimal lag selection in forecasting.

3. METHODOLOGY

3.1 Data Collection

Data collected was obtained from the database of Food and Agricultural Organization (FAO)[9], where the price value is Produce Price Index and the unit is in terms of Local Currency Unit (LCU) per tonne or Ringgit Malaysia (RM) per tonne. This dataset

consists of monthly price details of 31 commodities that starts from 1 January 2010.

3.2 Qualitative Study on Each Obtained Variables

3.2.1. Stationarity:

In time series analysis, a common assumption is that the data is stationary. It means the statistical properties of times series such as mean, variance and auto-correlation remains unchanged over time. In other words, time series that contains trends or seasonal components are not stationary. There are two ways to check for the stationarity in a time series:

1. Looking at Auto-correlation Function (ACF) plots. If the ACF plots looks exponentially decrease or decay as lags increases , this indicates as non-stationary.

2. Augmented Dicker Fuller Test. Null hypothesis for this test is that there is a unit root while the alternative hypothesis represents that no unit root in the time series. The null hypothesis can be rejected if the test statistic is less than the critical value, it means the time series is stationary. However, if the test statistic is larger than the critical value, the null hypothesis is failed to be rejected, it indicates that the time series is non stationary.

3.2.2. Difference Transform

When the time series is con-firmed to be non-stationary, but forecasting models always assume stationary series, there is a need to transform the series into stationary. Difference is one of the commonly used methods to reduce or even remove the trend and seasonality components. It aims to stabilize the mean of time series by removing the temporal dependence. This method is per-formed by the subtraction of previous observation (y_{t-1}) and current observation (y_t). In practice, a second or higher order difference may be needed until all the temporal structures in time series are removed.

$$\text{Difference}(t) = y_t - y_{t-1}$$

3.3 Optimal Lag Time Selection Function

As a population-based computation strategy, differential evolution contains 4 main operators: initialization, mutation, recombination and selection.

1) Initialization

At first, a population of possible sets of lags has been randomly generated. These individual has been represented as vector of lags weight with length equal to a predefined maximum lag allowed l_{max} . The initial value in each dimension of chromosome vectors was assigned randomly based on uniform distribution. This aim of distributing the values evenly is to maximise the exploration capabilities of the chromosome vectors in search space.

2) Mutation

The purpose of this operation is to calculate differentiation within population based on the distance and direction of lags values. According technique, the offspring can be considered as the adjustment of searching vector through alternative differentiation's direction with calculated step length rather than inherit character from parent's value. For each individual, the mutation vector will be defined by re-scaling the target with a predefined weight factor and adding a difference vector. The obtained vector is known as donor vector while the weight is notated as mutation factor.

3) Recombination

New trial vectors will be produced by incorporating with fittest individuals in previous generation. The elements of these trial vectors are taken from either donor vector or target vector respectively. Based on the purpose of evaluation, there are two different methods can be used to determine the new trial vector in a generation, namely binomial and exponential crossover.

4) Reselection

At this stage, target vector is compared against the trial vector in terms of fitness. If the newly generated vector results in higher fitness values than it parents, replacement will occur.

3.4 Artificial Neural Networks (ANNs)

With time series data, the nodes in input layer will be the lagged observations of the time series. In this research, a one-output neural network with single hidden layer is used for learning process and all the nodes in layers are fully connected. Simple back-propagation with gradient descend technique is involved for adjusting learning coefficient.

- **Input layer.** Given a maximum lags during DE operation, the number of input nodes will be the total number of

1. t number of past values in the series
2. t number of past values that is squared and scaled with correspondent coefficients
3. the current evaluation in term of time differences
4. a bias node.

- **Hidden layer.** The number of hidden nodes should be the number of input nodes by half.

- **Output layer.** One node in the output produces the end-price forecasting.

Steps involved in the hybrid model, which is the neural net-work based on differential evolution are given below:

Step 1. Initialize list of binary chromosome and execute DE process.

Step 2. Return DE output will be used to determine lag features that become input node in network. Initialize network structure.

Step 3. Generate chromosome and replace with the weight in network structure.

Step 4. Execute the DE process to adjust the weights, the objective function is to minimize RSME function.

Step 5. Final weight in DE process will be initialized for the network and start training process with back-propagation retrain.

Step 6. Validate the learning by cross-validate with 20% of instances from training set.

Step 7. Retrain step 5 if the learning has not validated or reach the maximum iteration.

Step 8. Output the final model and be available for forecasting with test set.

4. PRELIMINARY EXPERIMENTS AND RESULTS

4.1 Data Pre-processing

The asparagus data consists of attributes that needs to be removed, aggregate price by date. As the observations is in a monthly basis, the start of every month is used as timestamp. After the time series data is indexed using date-time objects, the data is now easy to use and can be visualised in a time series decomposition plot. From the figure 1, the price of asparagus is not stable with obvious trend and seasonal pattern. In other words, time series of asparagus is non-stationary, judging by the plots.

4.2 Auto-correlation

ACF plot can be done by using the *plot_acf()* function from the *statsmodels* library. From figure 2, the blue shaded region is the confidence level, the vertical lines are corresponding to each lag and the height represents the auto-correlation values for particular lag. Auto-correlation with lag zero is always equal to 1, because price and price with lag zero are the same variable. Starting from lag 1, it is obvious that there are some spikes go beyond the confidence interval region, which means the values are considered as statically significant. Auto-correlation values from lag 1 to lag 4 are statistically significant while spikes from lag 5 up to lag 18 are within the confidence level. This means the asparagus price is correlated with each other when the price is rising.

4.3 Stationarity Test

The test is implemented by using *adfuller()* provided by the *statsmodels* library. Null hypothesis for this test is that there is a unit root while the alternative hypothesis represents that no unit root in the time series.

In figure 3, the test statistic is -2.202397 and it is clearly shown that all the critical values are less than test statistic, which implies that the time series of asparagus price is not stationary. This confirms our previous observation based on the time series plot and auto-correlation plot.

4.4 Difference Transform

Pandas library provides a function *diff()* to calculate the difference of a time series dataset. This *diff()* function will compute first-order difference by default. Here, this function is used to perform the first order difference on time series of asparagus price. Meanwhile, auto-correlation plot for the first order difference time series is analysed. From the time series plot in Figure 4, first order difference successfully much of the trend variations in the time series. Looking at the auto-correlation plot, there is one spike go beyond the confidence interval region. Hence, a second order difference is carried out but the ACF plot is very similar to first order difference.

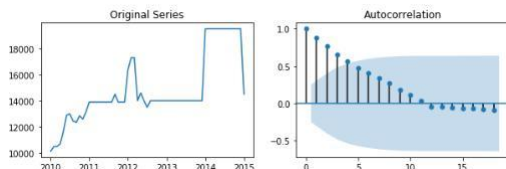


Figure 1. Time Series Decomposition Plot for Asparagus Price

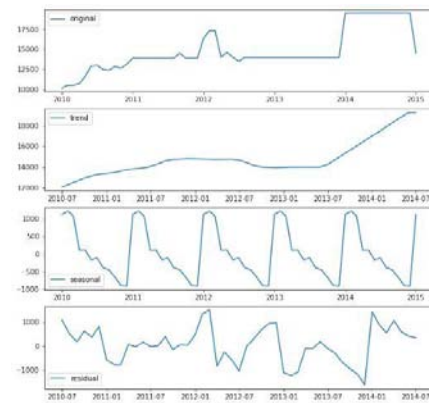
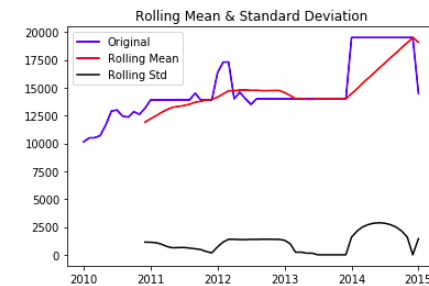


Figure 2. Auto-correlation plot for asparagus price.



```
Results of Augmented Dickey-Fuller Test:
Test Statistic      -2.202397
p-value             0.205377
#Lags Used          0.000000
Number of Observations Used  60.000000
Critical Value (1%)   -3.544369
Critical Value (5%)  -2.911073
Critical Value (10%) -2.593190
dtype: float64
```

Figure 3. ADF test for asparagus price.

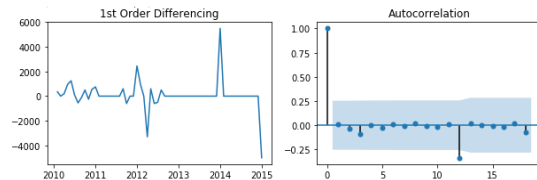


Figure 4. Auto-correlation for 1st order difference.

5. CONCLUSION

In summary, the decomposition plot depicts clearly that asparagus price is not stable, and consists of trend and seasonal components. From auto-correlation plot, there exists some auto-correlation values are statistically significant. Hence, asparagus price data are considered as non stationary. Results from ADF test provide evidence to support that the time series is non-stationary. First order difference is able to transform the series into stationary but further difference is not necessary.

Identifying the linkages between determinants and agricultural commodities prices in Malaysia is indeed crucial for policy makers and farmers. Data analysis and forecasting system can complement policy makers to formulate policy aiming at price stability by reducing the linkage between exogenous factors and the price. Similarly, with these information, farmers can immediately identify and address the issues found before huge losses. Lastly, it is advantageous for all the parties involved to pay attention at exogenous factors as well as the forecast result of commodities

price, and take immediate action such as adopt marketing strategies to manage the potential risk effectively.

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